Zhaozheng Shen

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Research Interests

Learning-based control and motion planning for robotic systems, with interests in reinforcement learning, model-based methods, and robotic manipulation. Applications include construction robotics and autonomous equipment.

Education

National University of Singapore Singapore
PhD in Civil and Environmental Engineering *Aug. 2025 – Present*
- Advisor: Assoc. Prof. Justin Yeoh

Tongji University Shanghai, China
Master of Engineering in Civil Engineering *Sep. 2022 – May 2025*
- Advisor: Prof. Jie Wu. Thesis: Construction Scheduling for Steel-Frame Structures Using Intelligent Algorithms

Tongji University Shanghai, China
Bachelor of Engineering in Civil Engineering *Sep. 2018 – Jul. 2022*

Publications

Journal Articles

- **Shen, Zhaozheng**; Wu, Jie. Multiobjective Ant Colony System Algorithm for Component-Level Construction Schedule Optimization. *Journal of Construction Engineering and Management*, 151(3), 04025002, 2025. doi:10.1061/JCEMD4.COENG-15827
- **Shen, Zhaozheng**; Wu, Jie; Cao, Yilun. A dual-adaptive directed genetic algorithm for construction scheduling. *Journal of Building Engineering*, 96, 110659, 2024. doi:10.1016/j.jobe.2024.110659

Conference Paper

- Jiang, Shanshan; Wang, Yifan; Yang, Bin; **Shen, Zhaozheng**. Reinforcement Learning–Driven Intelligent Path Planning for Dual Tower Cranes in Modular Construction Sites. In *Proc. 43rd Int. Symp. on Automation and Robotics in Construction (ISARC)*, pp. 634–641, 2026. doi:10.22260/ISARC2026/0082

Research Experience

Learning-Based Control of Underactuated Tower Cranes Mar. 2026 – Present
National University of Singapore, Advisor: Assoc. Prof. Justin Yeoh
- Built a Unity-based simulation and training environment for an underactuated tower crane, with actuated slewing, trolley travel, and hoisting coupled to passive payload swing and wind disturbances.
- Investigated reinforcement-learning policies for three-axis acceleration control, targeting rapid load transport, active braking, and suppression of residual payload swing at the destination.
- Benchmarked learned policies against anti-sway PID and nonlinear MPC using transport time, payload swing, endpoint overshoot, and residual motion, while investigating obstacle-aware extensions.

Robotic Interior Finishing for PPVC Modules Aug. 2025 – Present
Research Engineer, National University of Singapore (BCA-funded project)
- Developed Isaac Sim simulations and robotic operation sequences for tile installation and floor screeding in prefabricated modules.
- Processed floor point clouds to estimate reference planes, local slopes, and surface geometry for robotic floor-finishing planning.
- Studied wall spraying, tiling, and floor-finishing workflows through site investigations and task-time analysis to guide automation design.

Research Experience (continued)

Reinforcement Learning for Tower-Crane Selection and Site Layout Sep. 2026 – Present
National University of Singapore

- Investigating policy-gradient reinforcement learning for sequential tower-crane selection and temporary-facility placement under geometric and operational constraints.

Dual-Tower-Crane Path Planning in Modular Construction Oct. 2024 – Jun. 2025
Tongji University, Shanghai, China

- Contributed to reinforcement-learning reward design and curriculum training for cooperative crane path planning in a BIM-based Unity simulation (ISARC 2026).

Visual Next-Step Prediction in Robotic Disassembly Jul. 2024 – Present
Research Collaboration, Northeastern University (Remote), Advisor: Prof. Yang Ye

- Annotated over 1,000 robot-vacuum disassembly images, trained YOLOv8 component detectors, and developed RIVET, a multi-frame Transformer for next-action prediction from component states.
- Evaluated generalization to unseen robot units using robot-wise five-fold cross-validation; RIVET outperformed a frame-level YOLO-CLS baseline in accuracy and macro-F1.

Intelligent Construction Scheduling for Steel-Frame Structures Oct. 2022 – May 2025
Master's Thesis Research, Tongji University, Advisor: Prof. Jie Wu

- Derived 2,765 precedence constraints for a 1,165-member steel frame from IFC geometry and stability rules; developed a dual-adaptive directed GA for installation sequencing (*J. Build. Eng.* 2024).
- Developed a multiobjective ant colony system for multi-skill crew scheduling, optimizing duration, cost, and carbon emissions; benchmarked against NSGA-II and MOPSO (*ASCE JCEM* 2025).
- Investigated Dueling Double DQN for rescheduling under crew absences, late deliveries, and equipment failures.

Selected Course Projects

Robotics Course Projects (ME5402) 2026
National University of Singapore

- Modelled a wheel-legged robot limb as a planar 3R chain and designed a gravity-compensated joint-space PID controller using analytical kinematics and dynamics (individual project).
- Reproduced deep Q-learning for UR5 vision-based sorting in CoppeliaSim (group project), with an independent extension to MuJoCo grasp-and-carry with online replanning.

Teaching and Service

Student Helper, 43rd International Symposium on Automation and Robotics in Construction (ISARC) 2026
Graduate Office Assistant, Graduate School, Tongji University Sep. 2023 – Feb. 2024
Volunteer, Second National Intelligent Construction Academic Conference Dec. 2023
Teaching Assistant, Engineering Drawing (41 students), Tongji University Sep. 2022 – Feb. 2023

Industry Experience

Construction Management Intern, Shanghai Mechanized Construction Group Co. Ltd Jul. – Aug. 2021

Awards

Graduate Student Academic Scholarship, Tongji University 2024, 2023, 2022
Third Prize, Undergraduate Excellent Student Scholarship, Tongji University 2021

Skills

Learning and control: Reinforcement learning, imitation learning; PID, nonlinear MPC, motion planning
Simulation and perception: Unity, Isaac Sim, MuJoCo, CoppeliaSim; YOLO, Transformers, point clouds
Optimization: Evolutionary algorithms, nonlinear programming, multiobjective optimization
Programming and BIM: Python, MATLAB, JavaScript, Java; Neo4j, MongoDB; IFC, Revit, Tekla, AutoCAD